

Assignment 4.1 Feature Extraction Exercise

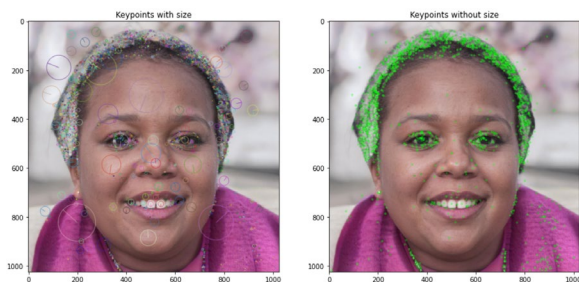
Part 1 - Feature Extraction

Start with reading the image shared with on BB (pic1 @thispersondoesnotexist). Read the image using OpenCV and create two versions of it. One colorful and one gray-scaled one. Store the Gray scaled image as “pic1_gray.jpg”. We need this image down the road.

Hint: By default the image may be loaded as BGR and you have to modify it to make it RGB



a) SIFT: In 2004, David Lowe introduced Scale-Invariant Feature transform as a method for extracting distinctive invariant features from images. The idea was feature matching between images. The biggest advantage of the method is detecting image features irrespective of orientation and size. For this part, please provide the SIFT using OpenCV on colorful image. Create two outputs one with sizing and one without it.



b) FAST: In 2006, Edward Rosten and Tom Drummond proposed “Features from accelerated segment test” (FAST) which is a corner detection method to extract feature points. This is especially popular in real-time video processing. OpenCV allows you to create this with or without suppression. Create two outputs one with max suppression and one without it.



c) ORB: Oriented FAST and Rotated BRIEF (ORB) was originally proposed by Ethan Rublee, Vincent Rabaud, Kurt Konolige, and Gary R. Bradski in 2011 (OpenCV Labs). The biggest advantage of this method is its robustness to distortion and noise.



d) One of the aspects of using feature extraction in OpenCV is to find feature matching between two images. Use this ability to make comparisons between the colorful and gray image. Print the number of matching points.

Part 2 - Face Recognition

For this part, we want to see how computer vision can help with face recognition (face detection). Start with downloading the image of Assignment_pic2.jpg file (@unsplash.com- a free resource of images). We need to start with Haar feature detection concept and use Haar cascading procedure. You can get more information about haar cascading [here](#). Download the required .xml files from assignment package.

Note: Using `cv2.imshow` is a tool for showing images in Python. However, this command makes crashes in Google colab. Instead use `cv2_imshow`. You need to call the package using: `from google.colab.patches import cv2_imshow`

- a) Apply the face and eye detection algorithm on the provided image.
- b) Connect your own webcam and see how accurately this can detect you in the image. To make it more challenging, you can bring in other people, animals, objects,.. in screen.

Part 3 - Parameter Detection

For the last part of the assignment, create a model that can count the number of apples in the image provided in Canvas ('apple.jpg'). For more information about the procedure, reference your *Practical Machine Learning for Computer Vision (Lakshmanan, 2021)* textbook. Your model should print the proper number of apples on the image.